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15 April 1954

MILITARY SPECIFICATION

VACUUM PUMP, ROTARY, POWER DRIVEN, 12 CFM, TYPE MA-1

1. SCOPE

1.1 This specification covers one type of 12-cfm, gasoline-engine-driven, skid-mounted, rotary vacuum pump, designated Type MA-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-Z-325	Zinc Plating (Electrodeposited)
TT-E-508	Enamel; Interior, Semigloss, Tints and White
TT-P-636	Primer Coating, Synthetic, Wood and Ferrous Metal
TT-T-656	Tricresyl Phosphate
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Military

JAN-P-100	Packaging and Packing For Overseas Shipment, General Specification
MIL-P-116	Preservation, Methods Of
MIL-C-490	Cleaning and Preparation Of Ferrous and Zinc Coated Surfaces For Organic Protective Coatings
MIL-L-2104	Lubricating Oil, Internal-Combustion Engine, Heavy-Duty
MIL-G-3056	Gasoline, Automotive, Combat
MIL-M-3171	Magnesium Alloy, Processes For Corrosion Protection Of
MIL-H-4797	Heater, Utility, Type F-2
MIL-S-5002	Surface Treatments (Except Priming and Painting) For Metal and Metal Parts in Aircraft

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MIL-D-5028	Drawings and Data Lists: Preparation Of Manufacturers (For Production Aeronautical And Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-5541	Chemical Films For Aluminum And Aluminum Alloys
MIL-E-5557	Enamel; Heat-Resisting, Glyceryl-Phthalate, Black
MIL-C-6529	Corrosion-Preventive Compound, Aircraft Engine
MIL-I-6866	Inspection, Penetrant Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-P-6889	Primer; Zinc-Chromate, For Aircraft Use
MIL-E-7729	Enamel, Gloss, Aircraft Application
MIL-A-8421	Air Transportability Requirements, General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports; Preparation Of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-E-11275	Engines, Gasoline, Industrial Type, 300 BHP and Under, General Specification
MIL-C-16232	Coatings, Phosphate, Heavy (Manganese And Zinc Type) And Phosphate Treating Solutions
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, For Steel Surfaces

STANDARDS

Federal

595	Colors

Military

MIL-STD-129	Marking For Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MS33586	Metals, Definition of Dissimilar

Air Force-Navy Aeronautical

AN6235	Filter Element - Hydraulic Replaceable Micronic Line Type
AN6236	Filter Element - Hydraulic Replaceable Micronic Reservoir Type

PUBLICATION

Air Force-Navy Aeronautical Bulletin

143	Specifications and Standards; Use of

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(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 OTHER PUBLICATION. - The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society of Mechanical Engineers

Power Test Code

19.5:4 Supplement on Instruments and Apparatus, part 5, chapter 4, Flow Measurement by Means of Standardized Nozzles and Orifice Plates.

(Copies of the ASME power test code may be obtained from the American Society of Mechanical Engineers, 29 West 39th Street, New York 18, New York.)

3. REQUIREMENTS

3.1 PREPRODUCTION. - This specification makes provisions for preproduction testing.

3.2 COMPONENTS. - The vacuum pump shall consist of the following major components:

DESCRIPTION	SEE REQUIREMENT
a. Power unit	3.7
b. Vacuum pump	3.8
c. Servicing line	3.9
d. Cabinet	3.10
e. Skid	3.11
f. Accessories	3.12

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS. - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143 except as specified in 3.3.1 and 3.3.2.

3.3.1 COMMERCIAL PARTS. - Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts.

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In any case, commercial utility parts, such as screws, bolts nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings

3.3.2 STANDARD PARTS. - With the exception specified in 3.3.1, AN and MS standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 MATERIALS

3.4.1 PROTECTIVE TREATMENT. - When materials are used in the contruction of the vacuum pump that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4.2 FUNGUS RESISTANCE. - As far as practicable, the vacuum pump shall be constructed of materials which are resistant to the deteriorating effects of fungus or treated in a manner which will adequately prevent fungus attack.

3.4.3 GASKETING AND INSULATING MATERIALS. - Rubber or similar gasketing and insulating materials shall be compounded to insure their oil resistance and ability to withstand extremely low temperatures.

3.4.4 METALS. - Wherever practicable, lightweight metals shall be used in the construction of the vacuum pump.

3.4.4.1 DISSIMILAR METALS. - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.5 COMBUSTIBLE MATERIALS. - Combustible materials shall be avoided in internal construction where high-purity oxygen (approximately 99.5 percent) may make contact during normal operation.

3.5 DESIGN AND CONSTRUCTION. - The vacuum pump shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 ADJUSTMENTS, REPAIRS, AND OVERHAUL. - The vacuum pump shall be constructed so that adjustments, repairs, and overhaul can be easily accomplished

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by the personnel of operating units and overhaul bases. The construction of the vacuum pump shall enable the performance of servicing and maintenance operations with standard handtools to the maximum practicable extent.

3.5.2 The vacuum pump shall be designed for the maximum practicable life between overhauls and to minimize loss of service time for maintenance, repair, and periodic servicing.

3.5.3 The vacuum pump shall be an independent, self-contained unit designed for the purpose of evacuating the vacuum-tight annular insulating space of liquid-oxygen or nitrogen storage tanks to absolute pressures of 10 microns Hg, or less.

3.5.4 LUBRICATION. - The vacuum pump shall be designed to perform efficiently and satisfactorily when operated with the lubricants specified herein. It shall be designed for economy in consumption of these lubricants.

3.5.5 PROTECTIVE COVERS. - Cover plates, gaskets, and fittings necessary for the protection of contained apparatus during operation, storage, or shipment shall be provided.

3.5.6 WINTERIZATION. - The vacuum pump shall be designed for starting and operating at any temperature between -65° and +130°F without external aid other than the use of a Type F-2 heater conforming to MIL-H-4797 during starting.

3.5.6.1 OIL DILUTION. - Oil dilution for starting shall not be acceptable.

3.5.7 HEATER CONNECTION. - When use of the Type F-2 heater is necessary, openings and fittings for attachment to the vacuum pump shall be provided in the enclosing cabinet. Removable covers shall be furnished for such openings. Covers shall be securely attached to the cabinet to prevent loss.

3.5.8 MOUNTING. - The vacuum pump shall be mounted on a sturdy skid and enclosed in a cabinet suitable for providing adequate protection from environmental conditions. Insofar as practicable, all individual components shall be located and installed on the skid to permit their being removed without removing other components.

3.5.9 LOCKING DEVICES. - All screws, pins, bolts, et cetera, shall be equipped with locking devices. Safety wire, self-locking nuts, cotter pins, lockwashers, et cetera, shall be acceptable.

3.5.10 AIR TRANSPORTABILITY. - The complete, assembled pump shall conform to MIL-A-8421.

3.5.11 SAFETY. - Generally accepted conservative engineering practice shall be used in the design and construction of the vacuum pump to insure adequate safeguards for operating personnel.

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3.5.11.1 GUARDS. - Rotating or reciprocating parts and dangerously hot surfaces that are so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or properly guarded.

3.5.12 CONTROLS. - The vacuum pump shall be equipped with all controls necessary to permit ready starting, stopping, and speed regulation.

3.6. PERFORMANCE. - The vacuum pump shall be capable of meeting the requirements specified herein.

3.6.1 CAPACITY

3.6.1.1 FREE AIR PUMPING. - The free air capacity of the vacuum pump shall be such that when it is operating at rated speed with both inlet and discharge connections open to atmospheric pressure, the quantity of air or other gases pumped through shall be in excess of 12 cfm measured at ambient temperature and pressure.

3.6.1.2 REDUCED PRESSURE. - The capacity of the vacuum pump at reduced pressures shall be such that when it is operating at rated speed in an ambient temperature of +70° F with the inlet pressure constant at 100 microns Hg, and the discharge open to atmospheric pressure, the quantity of air or other gases pumped through shall be not less than 8 cfm measured at inlet pressure and temperature conditions.

3.6.1.3 ULTIMATE VACUUM. - When operating at rated speed in an ambient temperature of +70° F with the inlet completely blanked off except for a connection to an absolute pressure measuring device and the discharge open to atmospheric pressure, the vacuum pump shall be capable of attaining and holding an absolute pressure of less than 10 microns Hg.

3.6.2 OPERATING TIME. - The vacuum pump shall be capable of continuous operation under any of the conditions specified in 3.6.1 for periods of not less than 12 hours without requiring shutdown for servicing, maintenance, or repairs.

3.6.3 ENVIRONMENTAL CONDITIONS

3.6.3.1 TEMPERATURE OPERATION. - The vacuum pump shall be capable of operating in any ambient temperature between the limits of -65° and +130° F when winterized as specified herein.

3.6.3.2 TEMPERATURE EXPOSURE. - The vacuum pump shall be capable of withstanding, without damage, storage at any temperature from -80° to +160° F.

3.6.3.3 HUMIDITY. - The vacuum pump shall not be damaged by operation or storage in any humidity conditions up to 95 percent relative humidity at 130° F.

3.6.3.4 FUNGUS. - The vacuum pump shall not be damaged by exposure to moist fungus encountered at locations in tropical and subtropical climates.

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3.6.3.5 SALT AIR. - The vacuum pump shall not be damaged by exposure to or operation in an atmosphere containing salt-laden moisture.

3.6.3.6 RAIN, SLEET, AND SNOW. - The vacuum pump shall not be damaged by storage or operation under adverse rain, sleet, or snow.

3.7 POWER UNIT. - The power unit shall consist of a gasoline engine power source and a power transmission system, together with the controls, instruments, and accessories needed for proper and safe operation of all components.

3.7.1 ENGINE. - The power source shall be a single, air-cooled, four-stroke-cycle, lightweight gasoline engine qualified to MIL-E-11275.

3.7.1.1 SIZE. - The gasoline engine power source shall be a group I unit with not less than one and one-half minimum net continuous brake horsepower at rated speed.

3.7.1.2 INTAKE AIR FILTER. - The intake air filter shall be the heavy-duty, oil bath type.

3.7.1.3 FUEL TANK. - A fuel tank as specified in MIL-E-11275 shall be provided. Fuel tank capacity shall be adequate to provide fuel for not less than 12 hours of continuous operation under any of the conditions specified in 3.6.1 through 3.6.1.3.

3.7.2 POWER TRANSMISSION SYSTEM. - A power transmission system consisting of a clutch, drive pulley, driven pulley, V-belt, and belt-tightening arrangement or a direct-drive assembly consisting of a clutch, gear reduction unit, and flexible coupling shall be provided for applying engine power to the vacuum pump driveshaft.

3.8 VACUUM PUMP. - The vacuum pump shall be of the single-stage, rotary, oil-sealed type. If used, the inlet valve shall be mechanically operated.

3.8.1 LUBRICANT. - The vacuum pump shall be designed to operate in accordance with the requirements specified herein when lubricated with commercial grade vacuum pump oil having a total vapor pressure of less than 1 micron Hg at +100° F.

3.8.1.1 The vacuum pump shall be designed so that it will operate, although not necessarily in compliance with all absolute pressure requirements specified herein, for extended periods when lubricated with type 2 tricresyl phosphate conforming to TT-T-656.

3.8.2 LUBRICATION SYSTEM. - The vacuum pump shall be provided with an automatic lubrication system which will insure an adequate lubricant supply to all parts requiring same under the operating conditions specified herein.

3.8.2.1 OIL CHECK DEVICE. - An automatic device shall be provided on each pump assembly which will prevent flow of the vacuum pump lubricant through the

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servicing line and into the container being evacuated in case the pump stops.

3.8.3 COOLING SYSTEM. - The vacuum pump shall have an air-cooling system with adequate capacity for maintaining the pump components at suitable operating temperatures under all environmental conditions specified herein.

3.8.4 SEPARATOR. - The vacuum pump shall incorporate features designed to effectively recover the lubricating fluid from the exhaust gases, separate condensable vapors from the lubricating fluid, and provide for discharging the condensable vapors from the pump. If a separator tank is provided, it shall be provided with a drain and drain shutoff valve.

3.8.5 INLET CONNECTION. - The vacuum pump inlet connection shall be of 1-inch iron pipe size or larger. A fitting suitable for vacuum-tight connection of the servicing line filter specified in 3.9.3 shall be provided in the pump inlet.

3.8.6 TACHOMETER. - The vacuum pump shall be equipped with a tachometer to indicate the main driveshaft speed. The tachometer dial shall be graduated from zero to not less than 125 percent of the maximum recommended pump speed. The tachometer shall be redlined at the maximum recommended speed.

3.9 SERVICING LINE. - A servicing line consisting of a length of vacuum line and a vacuum flange shall be provided with each vacuum pump.

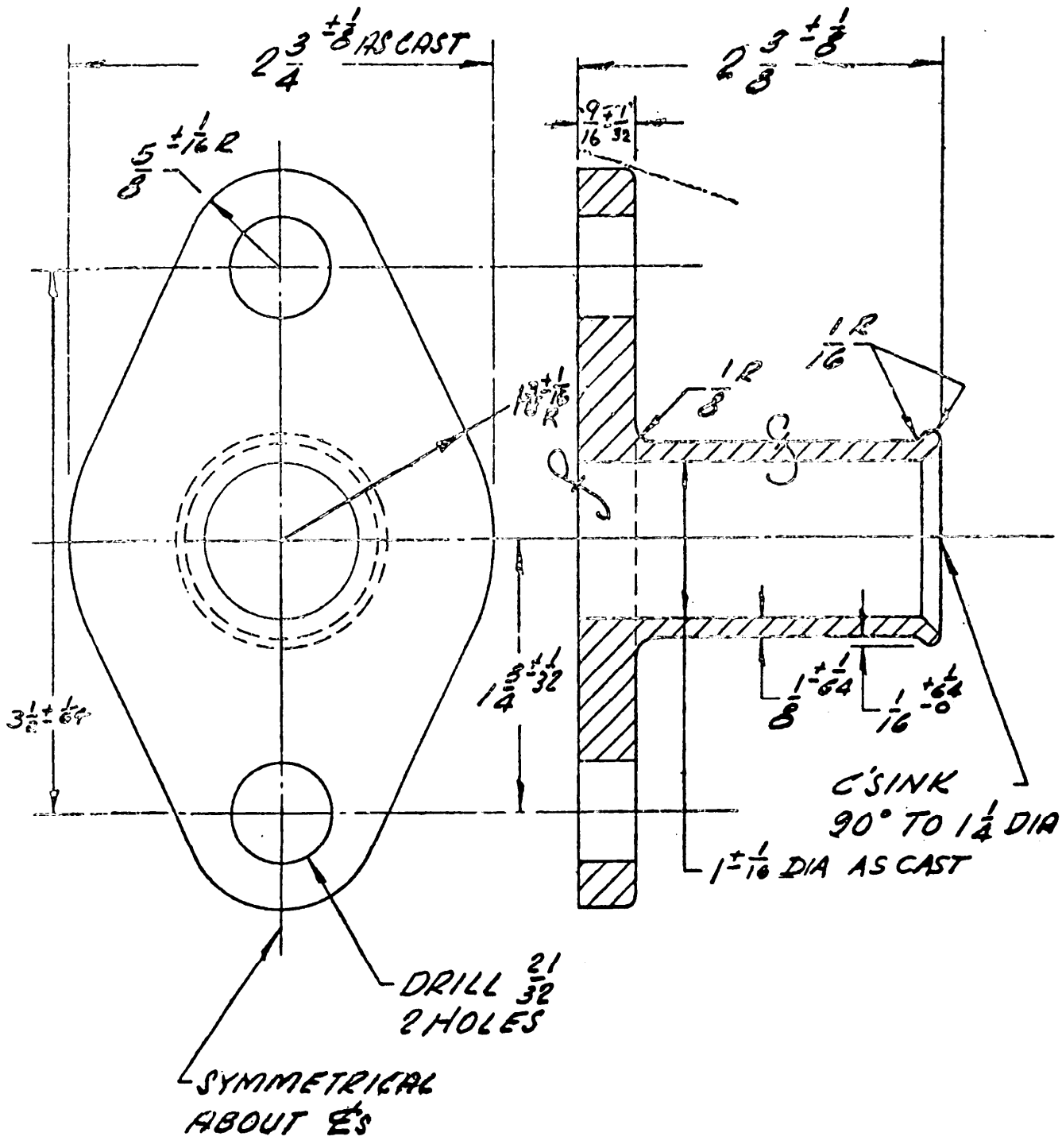
3.9.1 VACUUM LINE. - The vacuum line shall be a 72 ±6-inch section of heavy-duty, flexible, 1-inch internal diameter vacuum hose designed for use at absolute pressures of 10 microns Hg and below. One end of the vacuum line shall be prepared as necessary for connection to the servicing line filter inlet fitting. If necessary, a metal band clamp shall be provided to prevent the hose from being pulled off the fitting during use.

3.9.2 VACUUM FLANGE. - A vacuum flange conforming to figure 1 shall be provided. The vacuum flange shall be connected to the vacuum line in a manner that will insure against leaks. If necessary, a metal band clamp shall be provided to prevent the hose from being pulled off the flange during use.

3.9.3 SERVICING LINE FILTER. - The vacuum pump shall be equipped with a replaceable-element-type filter between the servicing line and vacuum pump inlet to prevent passage of insulating material into the vacuum pump from the liquid-oxygen storage tank annular space being evacuated.

3.9.3.1 GENERAL FILTER DESIGN. - The servicing line filter shall be designed for solid and secure installation within the vacuum pump enclosing cabinet, so that it can be readily opened for inspection and replacement of elements, and so that it will seal vacuumtight upon reassembly. A fitting suitable for vacuum-tight connection of the servicing line specified in 3.9 shall be provided in the servicing line filter element.

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DIMENSIONS IN INCHES.

SMOOTH MACHINE FINISH WHERE SHOWN ℄

VACUUM FLANGE

FIGURE 1

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3.9.3.2 FILTER SIZE. - The servicing line filter shall be of sufficient size and shall use a sufficient number of individual elements to insure free flow of gases into the vacuum pump from the container being evacuated.

3.9.3.3 FILTER ELEMENTS. - The servicing line filter shall be designed to use replaceable elements conforming to AN6235 or AN6236.

3.10 CABINET. - A sturdy cabinet for containing and providing protection for the vacuum pump components, accessories, controls, and instruments shall be fabricated onto the skid base. The cabinet shall be constructed so that it, or applicable sections, can be readily removed from the vacuum pump to facilitate maintenance on or removal of components.

3.10.1 The cabinet shall be designed and constructed to protect the items installed within from adverse weather conditions. The top of the cabinet and the tops of all protuberances therefrom shall be shaped and slanted to resist the retention of rain, snow, or sleet. Panels designed to open for operating the vacuum pump shall be given particular attention in this respect so that snow or ice cannot lodge thereon in such a manner as to prevent ready access or actuation.

3.10.2 SIZE. - The cabinet shall be of sufficient size to provide access to vacuum pump components for maintenance and repair. Space shall be provided for storing the servicing line within the cabinet when not in use.

3.10.3 ACCESS DOORS. - The cabinet shall include sufficient access doors to provide adequate ventilation during operation and to provide ready access to all components for maintenance or repair while they are installed in the cabinet.

3.10.3.1 All access doors shall be equipped with continuous hinges on one edge and shall be provided with latches for holding them securely in the closed position. Means shall be provided for securing them in the open position if such is necessary during operation. Access door latches shall be constructed to permit operation by personnel wearing heavy gloves.

3.11 SKID. - The vacuum pump shall be solidly mounted on a lightweight metal skid base. The skid base shall be of sufficient dimensions to forestall any tendency of the vacuum pump to overturn, and to permit operation of the vacuum pump on any flat, reasonably smooth, level surface without damage or excessive vibration.

3.12 ACCESSORIES

3.12.1 PROTECTIVE CAPS. - Protective caps shall be provided to prevent moisture and other foreign material from entering the vacuum pump or hose during storage and transport. If the servicing line must be removed from the vacuum pump inlet fitting for convenient storage within the cabinet, caps shall be provided for both ends of the hose and for the vacuum pump inlet fitting. If it will not be necessary to remove the line for storage, one cap to close the vacuum flange end of the servicing line shall be adequate.

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3.12.2 OPERATION AND MAINTENANCE TOOLS. - Sets of all special tools (other than common handtools and those tools normally available in motor shops) required for proper operation, maintenance, and repair of the vacuum pump shall be furnished (see 6.2).

3.12.3 TOOL COMPARTMENT. - When specified, a compartment suitable for storage of the special tools specified in 3.12.2 shall, if applicable, be provided within the cabinet (see 6.2).

3.12.4 HANDBOOK COMPARTMENT. - A handbook compartment shall be provided within the cabinet. The handbook compartment shall be located to be readily accessible.

3.13 INTERCHANGEABILITY. - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. The drawing number requirements of MIL-D-5028 shall govern changes in the manufacturer's part numbers.

3.14 DIMENSIONS. - The overall outside dimensions of the vacuum pump, including cabinet and skid base, shall not exceed 45 inches in length, 30 inches in width, and 30 inches in height.

3.15 WEIGHT. - The weight of the vacuum pump shall not exceed 350 pounds.

3.16 FINISHES AND PROTECTIVE COATINGS

3.16.1 EXPOSED PARTS AND SURFACES. - All exposed metal and surfaces inside or outside the cabinet, with the exception of certain corrosion-resistant materials installed within the cabinet, shall be treated as specified herein.

3.16.1.1 CORROSION-RESISTANT METAL PARTS AND SURFACES. - Metal parts and surfaces that are installed within the cabinet and are fabricated from corrosion-resistant materials need not be given protective coatings if such coatings will interfere with their proper functioning.

3.16.2 CHEMICAL TREATMENT AND PRIMING

3.16.2.1 FERROUS MATERIALS. - Ferrous parts or surfaces shall be cleaned and given a chemical treatment in accordance with MIL-C-490. A primer coat conforming to TT-P-636 or MIL-P-26915 shall be applied.

3.16.2.1.1 All exposed ferrous parts such as bolts, screws, nuts, washers, et cetera, shall be cadmium plated in accordance with QQ-P-416, given a phosphate protective coating conforming to MIL-C-16232, or zinc plated in accordance with QQ-Z-325. Cadmium plating shall be type II or III, as applicable.

3.16.2.2 ALUMINUM MATERIALS. - Aluminum and aluminum-alloy parts and surfaces shall be cleaned in accordance with MIL-S-5002. A corrosion-preventive

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chemical film conforming to MIL-C-5541 followed by a primer coat conforming to MIL-P-6889 shall be applied. An anodic film conforming to MIL-A-8625 may be used in lieu of the chemical film in accordance with MIL-C-5541.

3.16.2.3 MAGNESIUM MATERIALS. - Magnesium base alloy parts shall be cleaned and chemically treated in accordance with MIL-M-3171. Two primer coats conforming to MIL-P-6889 shall be applied.

3.16.2.4 OTHER SURFACES. - Other types of surfaces shall be cleaned and chemically treated in accordance with MIL-T-704, followed by a primer coat conforming to MIL-P-6889.

3.16.3 FINISH COATS

3.16.3.1 ENCLOSING CABINET EXTERIOR. - Exterior surfaces and parts of the vacuum pump enclosing cabinet shall be finished with two coats of yellow-orange gloss enamel, color 506, conforming to MIL-E-7729.

3.16.3.2 ENCLOSING CABINET INTERIOR. - Interior surfaces and parts of the enclosing cabinet shall be finished with two coats of semigloss enamel conforming to TT-E-508. The color shall be light green, color 24533 of Standard 595.

3.16.3.3 EQUIPMENT. - Equipment installed within the enclosing cabinet, except rubber or composition rubber parts and those parts subject to high operating temperatures, shall be finished with two coats of aircraft gray gloss enamel, color 512, conforming to MIL-E-7729. Parts subject to high temperatures, such as engine cylinder heads, manifolds, shrouding, exhaust system parts, et cetera, shall be finished with two coats of heat-resistant black enamel conforming to MIL-E-5557.

3.17 OPERATING INSTRUCTIONS. - Brief operating and precautionary instructions shall be suitably and permanently affixed on or near the control panel. The instructions shall be clear, concise, and adequate to enable operation without damage to the equipment or danger to the operator. The instruction panel shall be of such composition that exposure to oil, dirt, light, et cetera, will not fade it or cause it to become eradicated.

3.18 IDENTIFICATION OF PRODUCT. - Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.19 WORKMANSHIP

3.19.1 GENERAL. - The vacuum pump, including all parts and accessories, shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

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3.19.2 DIMENSIONS. - Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the vacuum pump or any of its components, they shall be held or limited accordingly.

3.19.3 SCREW ASSEMBLIES. - Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.19.4 RIVETING. - Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.19.5 CLEANING. - The vacuum pump shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 CLASSIFICATION OF TESTS. - The inspection and testing of the vacuum pump shall be classified as follows:

a. Acceptance tests . . . . . See 4.2

b. Preproduction testing . . . . . See 4.5

4.2 ACCEPTANCE TESTS. - The acceptance tests shall consist of the following:

a. Individual tests

b. Periodic sampling plans and tests

4.2.1 INDIVIDUAL TESTS. - Each vacuum pump shall be subjected to the following tests as described under 4.4:

a. Examination of product . . . . . See 4.4.1

b. Mechanical inspection . . . . . See 4.4.2

c. Functional check . . . . . See 4.4.3

d. Operational check . . . . . See 4.4.4

4.2.2 PERIODIC SAMPLING PLANS AND TESTS

4.2.2.1 SAMPLING PLAN. - One vacuum pump shall be selected at random from every fifty or fraction thereof produced and subjected to the tests specified in 4.4.5.

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4.2.2.1.1 REJECTION AND RETEST. - When one vacuum pump selected from a production run fails to meet the specification, vacuum pumps still on hand or later produced shall not be accepted until the extent and cause of failure are determined.

4.2.2.1.2 INDIVIDUAL TESTS MAY CONTINUE. - For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of vacuum pumps on hand or produced later shall not be made until it is determined that the vacuum pumps meet all the requirements of the specification.

4.2.2.1.3 DEFECTS IN VACUUM PUMPS ALREADY ACCEPTED. - The investigation of a test failure could indicate that defects may exist in vacuum pumps already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 TEST CONDITIONS

4.3.1 APPARATUS. - Insofar as practicable, apparatus used in conjunction with testing prescribed herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.3.2 INSTRUMENTATION

4.3.2.1 PRESSURES. - Wherever practicable, pressures and pressure differentials shall be measured by means of liquid manometers. Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Such gages shall be of laboratory type having a dial diameter of not less than 4 inches.

4.3.2.1.1 Pressures in the range from 1 to 100 mm Hg absolute shall be measured with closed-type manometers providing readings independent of prevailing barometric conditions.

4.3.2.1.2 Absolute pressures below 1 mm Hg shall be measured by the use of suitably calibrated instruments designed to read total pressures (including both permanent gases and condensable vapors). A certificate of instrument calibration to show compliance with 4.3.3 through 4.3.3.8 shall be included as part of the test report specified in 4.5.2.

4.3.2.1.3 Barometric pressures shall be measured by means of a properly calibrated mercurial or aneroid barometer.

4.3.2.2 TEMPERATURES. - Temperatures shall be measured by means of appropriately located thermometers or thermocouples used with suitably calibrated potentiometers. Temperatures shall be expressed in degrees Fahrenheit.

4.3.2.3 AIRFLOW

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4.3.2.3.1 Airflow for free air displacement tests shall be measured with standardized nozzles or orifices as prescribed in ASME Code 19.5:4. The rate of airflow shall be reported in terms of the actual volume occupied under the existing ambient conditions.

4.3.2.3.2 When conducting tests to determine the pumping capacity under vacuum conditions, airflow into the evacuated container (if applicable) shall be measured by a method for which an error of less than 1 percent can be demonstrated. The rate of airflow under such conditions shall be reported in terms of the actual volume it would occupy under the pressure and temperature conditions existing in the evacuated container.

4.3.2.4 REVOLUTIONS PER MINUTE. - Whenever practicable, rotational speed shall be determined by means of a directly coupled positive counter which will actually count the revolutions for a period of not less than 1 minute. A combination unit which counts the revolutions while measuring the time required is preferred. Strobotachometers may be used provided calibration data can be furnished to attest to their accuracy. The use of hand tachometers shall not be acceptable.

4.3.3 TOLERANCES

4.3.3.1 Data on absolute pressures measured in the range from 0 to 10 microns Hg shall be accurate to within 2 microns Hg.

4.3.3.2 Data on absolute pressures measured in the range from 10 microns Hg to 50 microns Hg shall be accurate to within 5 microns Hg.

4.3.3.3 Data on absolute pressures measured in the range from 50 microns Hg to 100 microns Hg shall be accurate to within 10 microns Hg.

4.3.3.4 Data on abolute pressures measured in the range from 100 to 500 microns Hg shall be accurate to within 25 microns Hg.

4.3.3.5 Data on absolute pressures measured in the range from 500 microns to 1 mm Hg shall be accurate to within 50 microns Hg.

4.3.3.6 Data on absolute pressures measured in the range from 1 mm Hg to atmospheric pressure shall be accurate to within 0.1 mm Hg.

4.3.3.7 Data on airflow rates shall be accurate to within 1 percent.

4.3.3.8 Data on temperatures shall be accurate to within 0.5° F.

4.3.4 ENGINE FUEL. - The gasoline engine fuel to be used throughout the testing specified herein shall conform to MIL-G-3056.

4.3.4.1 FUEL CONSUMPTION. - Fuel consumption measurement shall be made by either the volume or the weight method, and shall be determined by measuring the

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quantity necessary to refill a reservoir after a suitable elapsed time, or by measuring the drop of fuel level over a suitable elapsed time in a container of accurately known dimensions. Regardless of the method used, all expressions of quantities shall be recorded on a volume basis.

4.3.5 ENGINE LUBE OIL. - The gasoline engine lube oil to be used throughout the testing specified herein shall conform to MIL-O-2104.

4.3.5.1 LUBE OIL CONSUMPTION. - Engine oil consumption shall be determined by measuring the quantity necessary to maintain a given level in the oil reservoir. Oil consumption shall be recorded on a volume basis.

4.3.6 VACUUM PUMP LUBRICANT. - Commercial grade vacuum pump oil having a total vapor pressure of less than 1 micron Hg at +100° F shall be used as the vacuum pump lubricant and sealing fluid in all performance tests except as follows:

4.3.6.1 Type 2 tricresyl phosphate conforming to TT-T-656 shall be used as the vacuum pump lubricant and sealing fluid in the tests specified in 4.4.8.4 and 4.4.9.

4.3.6.2 VACUUM PUMP LUBRICANT CONSUMPTION. - Vacuum pump lubricant consumption shall be determined by measuring the quantity necessary to maintain a given level in the oil reservoir. Lubricant consumption shall be recorded on a volume basis.

4.3.7 INSPECTION, SERVICING, AND ADJUSTING. - Inspections, servicing, and adjustments shall be performed as specified by the manufacturer unless such directions conflict with the requirements of this specification. Throughout the endurance tests, these services shall be performed during the shutdown periods. Engine servicing shall follow that specified in MIL-E-11275 as closely as practicable.

4.3.8 OBSERVATIONS. - During the progress of all tests, such characteristics as smoothness of operation, vibrations, ease of control, and the operation of any auxiliary equipment shall be carefully observed to determine whether the vacuum pump complies with the requirements specified herein.

4.3.9 PREPARATION FOR TEST. - The vacuum pump shall be prepared for test by adjusting and servicing as specified by the manufacturer, except that the fuel and lubricating fluids specified herein shall be used.

4.3.10 TEST DATA

4.3.10.1 The following operational data shall be recorded at intervals not to exceed 15 minutes in performance checks and other short duration runs:

a. Vacuum pump speed, rpm

b. Absolute pressure of air or other gases entering the pump inlet, mm Hg or microns Hg, as applicable

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c. Temperature of air or other gases entering the pump inlet, degrees Fahrenheit

d. Airflow entering vacuum pump, expressed in cfm of air or other gases measured under the inlet temperature and pressure conditions

4.3.10.2 The barometric pressure and temperature of the ambient air shall be recorded at intervals not to exceed 1 hour in performance checks and other short duration runs, and at intervals not to exceed 4 hours in endurance test runs.

4.3.10.3 The following operational data shall be recorded at intervals not to exceed 1 hour throughout the endurance test operational runs:

a. Vacuum pump speed, rpm

b. Absolute pressure of air or other gases entering the pump inlet, mm Hg or microns Hg, as applicable

c. Temperature of air or other gases entering the pump inlet, degrees Fahrenheit.

4.3.10.4 Fuel and lubricant consumption data shall be reported at intervals not to exceed 24 hours in endurance test runs. Fuel and lubricant consumption shall be reported in terms of quarts consumed per hour of operation.

4.3.11 OPERATING LIMITS

4.3.11.1 AMBIENT TEMPERATURE. - Insofar as practical, performance and endurance testing shall be conducted within ambient temperature limits of from 50° to 100° F.

4.3.11.2 VACUUM PUMP SPEED. - The vacuum pump speed shall be maintained within ±2 percent of that recommended by the manufacturer.

4.3.12 TEST TIME. - Intervals of endurance test time less than one complete 12-hour operational period terminated by a parts failure shall not be credited to the endurance test. Endurance test time shall be credited in 12-hour increments.

4.3.13 RUNNING-IN. - The nature and extent of running-in shall be determined by the manufacturer. All necessary adjustments other than normal control adjustments shall be made during this run and shall remain undisturbed thereafter.

4.4 TEST METHODS

4.4.1 EXAMINATION OF PRODUCT. - The vacuum pump shall be inspected to determine compliance with the requirements specified herein with the respect to materials, workmanship, dimensions, and marking.

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4.4.2 MECHANICAL INSPECTION. - A mechanical inspection shall be made of all components and parts of the vacuum pump. All pertinent data concerning condition, defects of manufacture, damage in transit, and damage through use prior to test shall be recorded.

4.4.3 FUNCTIONAL CHECK. - All mechanical parts of the vacuum pump shall be carefully checked for free and proper functioning.

4.4.4 OPERATIONAL CHECK. - The vacuum pump shall be subjected to and shall satisfactorily complete the following tests before acceptance, and shall exhibit no malfunctions, leaks, excessive vibration, or other adverse operational characteristics.

4.4.4.1 INDIVIDUAL PROOF RUNNING. - The vacuum pump shall undergo an operational proof run of not less than 12 hours. The proof run time shall be divided as follows and in the order listed:

a. Four hours of continuous operation at rated speed with the vacuum pump inlet and discharge connections open to atmospheric pressure

b. Four hours of continuous operation at rated speed with the vacuum pump inlet pressure maintained between 50 and 500 microns Hg absolute

c. Four hours of continuous operation at rated speed with the vacuum pump inlet completely blanked off except for a connection to the absolute pressure measuring device. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation.

4.4.5 PERFORMANCE CHECKS. - The vacuum pump shall be subjected to the following performance checks:

a. Free air pumping rate (See 4.4.8.1)

b. Pumping rate at low pressures (See 4.4.8.3)

4.4.6 PRELIMINARY RUN-IN. - Any preliminary run-in specified by the manufacturer shall be performed prior to further testing.

4.4.7 PREPERFORMANCE MECHANICAL INSPECTION AND TEST

4.4.7.1 INITIAL TEARDOWN INSPECTION. - The vacuum pump used for testing shall be disassembled and inspected prior to further tests. All wearing parts and all parts subjected to high stresses in operation shall be accurately measured, ferrous parts shall be subjected to magnetic particle inspection in accordance with MIL-I-6868, and nonmagnetic parts shall be inspected with fluorescent penetrant (black light) in accordance with MIL-E-6866 for comparison with like inspections to be performed after completion of the tests. The vacuum pump shall be reassembled and subjected to the remainder of the tests.

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4.4.8 PERFORMANCE DEMONSTRATION. - The vacuum pump shall be tested in accordance with the following procedures to determine its perfomance capabilities prior to endurance testing. Any malfunctions or unusual operating difficulties occurring during the tests shall be recorded.

4.4.8.1 FREE AIR PUMPING RATE. - The vacuum pump shall be operated at rated speed until its free air pumping rate, the number of actual cubic feet measured at ambient conditions it will pump through per minute when both inlet and discharge are at atmospheric pressure, has been determined. The free air displacement shall be measured by means of a flowmeter connected to the inlet opening. The duration of this test shall be not less than 2 hours.

4.4.8.2 ULTIMATE VACUUM. - The vacuum pump inlet shall be blanked off except for a connection to the absolute pressure measuring device and operated at rated speed with the discharge open to atmospheric pressure until the inlet pressure has remained stable for a period of not less than 1/2 hour. A check shall be made to insure that no flow of air or other gases is entering the pump inlet during this test. The vacuum pump shall have operated continuously for not less than 1 hour prior to making the pressure check.

4.4.8.3 PUMPING RATE AT LOW PRESSURES. - The vacuum pump shall be operated at its rated speed with the discharge open to atmospheric pressure until the rate in cfm (measured at inlet conditions of pressure and temperature) at which it is capable of evacuating a container maintained at 100 microns Hg absolute pressure has been determined. This test shall be conducted by one of the following methods:

4.4.8.3.1 MEASURED LEAK METHOD. - The vacuum pump shall be connected to a vacuum chamber of suitable size which has been tested and found to have no detectable leaks when operated at an absolute pressure of 100 ± 10 microns Hg. The vacuum pump shall be started and operated to evacuate the chamber to approximately 100 microns Hg absolute pressure. Air shall be permitted to leak into the chamber through a flow measuring device at a rate which will maintain the chamber pressure constant at 100 ± 10 microns Hg absolute pressure. The quantity of air permitted to leak into the chamber over a period of not less than 2 hours shall be determined. The quantity measured shall be converted to the actual volume at chamber pressure and temperature that was permitted to leak into the chamber per minute (actual pumping rate). The volumetric efficiency of the vacuum pump when operating at an absolute pressure of 100 microns Hg shall be calculated. The method of calculation for determining both the pumping rate and the volumetric efficiency shall be demonstrated.

4.4.8.3.2 PRESSURE REDUCTION METHOD. - The vacuum pump shall be connected to a chamber of accurately known dimensions which has been tested and found to have no detectable leaks when operated at an absolute pressure of 100 ± 10 microns Hg. The vacuum pump shall be started and used to evacuate the chamber to a pressure less than 100 microns Hg absolute. The time required to evacuate the chamber from an absolute pressure of, for example, 200 to 50 microns Hg, shall be accurately determined by a series of not less than 10 individual tests. The internal volume of the

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vacuum chamber shall be not less than 10 cu ft. From the known internal volume of the chamber plus its attachments and the length of time required to reduce the absolute pressure as indicated above, the pumping rate at a pressure of 100 microns Hg absolute shall be determined. The volumetric efficiency of the vacuum pump at an absolute pressure of 100 microns shall be calculated from this information. The methods of calculation for determining the pumping rate and the volumetric efficiency shall be demonstrated.

4.4.8.3.3 The connection from the vacuum chamber to the vacuum pump shall be of 1-inch iron pipe size.

4.4.8.4 TEST WITH TRICRESYL PHOSPHATE. - The tests specified in 4.4.8.1, 4.4.8.2, and 4.4.8.3 shall be repeated using tricresyl phosphate as specified in 4.3.6.1 for the pump lubricant and sealing fluid. The tricresyl phosphate may be processed to remove water.

4.4.9 ENDURANCE. - The vacuum pump shall be subjected to an endurance test of not less than 504 hours. The lubricant and sealing fluid used throughout this test shall be tricresyl phosphate as specified in 4.3.6.1. The endurance test time shall be divided as follows:

a. One hundred and twenty hours of operation with the vacuum pump inlet and discharge connections open to atmospheric pressure

b. One hundred and twenty hours of operation with the vacuum pump inlet completely blanked off except for a connection to the absolute pressure measuring device and the discharge open to atmospheric pressure. A check shall be made to insure that no flow of air or other gases is entering the pump inlet during this test.

c. Two hundred and sixty four hours of operation with the vacuum pump inlet pressure maintained between 50 and 500 microns Hg absolute

4.4.9.1 TEST PROCEDURE. - The endurance test time shall be accumulated by continuous operation at rated speed for periods of 12 hours under the specified conditions followed by shutdown periods of not less than 6 hours. Inspections, lubrication, minor adjustments, minor repairs, and preventive maintenance shall be accomplished during the shutdown periods.

4.4.10 PERFORMANCE RECHECK. - Following the endurance test run and without intervening reconditioning other than normal preventive maintenance and servicing with lubricants, the vacuum pump shall be resubjected to the performance demonstration specified in 4.4.8. The performance at this time shall be not less than that specified in section 3 for a new vacuum pump.

4.4.11 FINAL TEARDOWN INSPECTION. - The vacuum pump shall be disassembled as specified in 4.4.7.1 and a critical inspection made of all components to determine their operability and any damage or undue wear encountered during the testing.

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4.4.11.1 The measurements taken under 4.4.7.1 shall be repeated for comparison with the data recorded prior to testing. Wear or distortion which exceeds the limits established by the manufacturer for new parts shall be cause for considering the affected parts as having failed to satisfactorily pass the tests.

4.4.11.2 Ferrous parts shall be resubjected to magnetic particle inspection (magnaflux) in accordance with MIL-I-6868 and shall exhibit no indications of damage attributable to the tests, or they shall be considered as having failed the tests. Grind checks and subsurface indications of laps, seams, or other forging defects noted before the test and which have not increased in size shall not be cause for rejection.

4.4.11.3 Nonmagnetic casting, forgings, and other operating parts shall be subjected to inspection with fluorescent penetrant (black light) in accordance with MIL-I-6866, and shall exhibit no cracks or other defects, or shall be considered to have failed the tests.

4.4.12 ENVIRONMENTAL TESTING. - The vacuum pump shall be subjected to the following environmental tests after the endurance tests or, if desired, at any time during the interval in which the endurance tests are made:

4.4.12.1 OPERATIONAL TESTS

4.4.12.1.1 HIGH TEMPERATURE START AND OPERATION. - The vacuum pump shall be exposed to an ambient temperature of 130° ± 5° F for not less than 36 hours. The vacuum pump shall be started and operated at rated speed for not less than 2 hours under this ambient temperature with the inlet blanked off except for the absolute pressure measuring device connection. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation.

4.4.12.1.2 ZERO START AND OPERATION. - The vacuum pump shall be exposed to an ambient temperature of 0° ± 5° F for not less than 36 hours. The vacuum pump shall be started in accordance with the manufacturer's instructions, but without use of the external heater specified in 3.5.6, and operated at rated speed for not less than 2 hours under this ambient temperature with the inlet blanked off except for the absolute pressure measuring device connection. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation.

4.4.12.1.3 LOW TEMPERATURE START AND OPERATION. - The vacuum pump shall be exposed to an ambient temperature of -65° ± 5° F for not less than 36 hours. The vacuum pump shall be heated and started in accordance with manufacturer's instructions and operated for not less than 2 hours under this ambient temperature with the inlet blanked off except for the absolute pressure measuring device connection. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation.

4.4.12.2 ATMOSPHERIC TESTS

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4.4.12.2.1 HIGH TEMPERATURE EXPOSURE. - The vacuum pump shall be exposed to, but not operated in, an ambient temperature of 160° ± 5° F for not less than 50 hours in accordance with procedure I of MIL-E-5272. Upon completion of this exposure, the vacuum pump shall be inspected for operability, distortion, cracking, differential expansion, pliability, and other detrimental effects on nonmetallic materials.

4.4.12.2.2 LOW TEMPERATURE EXPOSURE. - The vacuum pump shall be exposed to, but not operated in, an ambient temperature of -80° ±5° F for not less than 48 hours. Upon completion of this exposure, the vacuum pump shall be inspected for operability, distortion, cracking, differential contraction, embrittlement, and other detrimental effects on nonmetallic materials.

4.4.12.2.3 HUMIDITY. - The vacuum pump shall be exposed, but not operated, in accordance with humidity tests, procedure I of MIL-E-5272. The vacuum pump shall be carefully inspected for any damage caused by the high humidity and temperature, and operated at rated speed for not less than 2 hours under ambient atmospheric conditions with the inlet blanked off except for the absolute pressure measuring device connection. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation.

4.4.12.2.4 SALT. - The vacuum pump shall be exposed, but not operated, in accordance with the salt spray test, procedure I of MIL-E-5272. The vacuum pump shall be carefully inspected for any damage caused by the salt atmosphere, and operated at rated speed for not less than 2 hours under ambient atmospheric conditions with the inlet blanked off except for the absolute pressure measuring device connection. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation.

4.4.12.2.5 FUNGUS. - The vacuum pump shall be exposed, but not operated, in accordance with the fungus tests, procedure I of MIL-E-5272. The vacuum pump shall be checked carefully for fungus damage.

4.4.12.2.6 RAIN TEST. - The vacuum pump shall be exposed, but not operated, in accordance with the rain tests, procedure I of MIL-E-5272. Upon completion of the rain exposure, the vacuum pump shall be inspected to determine the effectiveness of the enclosing cabinet for protection against rain.

4.5 PREPRODUCTION TESTING

4.5.1 PREPRODUCTION TEST SAMPLES. - The preproduction test sample shall consist of one vacuum pump representative of the production equipment. It shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 PREPRODUCTION TEST REPORT. - If the testing is performed by the contractor, he shall prepare a preproduction test report in accordance with MIL-T-9107 and furnish three complete copies of the report to the procuring activity.

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4.5.3 PREPRODUCTION TESTS. - The preproduction tests shall consist of all the tests described under 4.4.

4.5.4 PART FAILURES. - If during the preproduction testing of a vacuum pump of an unapproved type a part fails, that part shall be replaced with a redesigned part or one of a different material unless the procuring activity authorizes installation of a new part of the original design and material for one which failed due to faulty material or workmanship. The preproduction testing shall be considered completed when every major part in the vacuum pump has passed the complete test. At the discretion of the procuring activity, redesigning and retesting may be required of any part which fails or indicates weakness after completing its preproduction test, but which is retained in the vacuum pump to complete testing of other parts.

4.5.5 MATERIALS AND INFORMATION. - When items are submitted for test, they shall be accompanied by the applicable bill of materials and manufacturer's drawings, complete operating instructions, and adequate spare gaskets, locking devices, seals, et cetera, to provide replacements which may be necessary for the disassemblies and reassemblies required during the testing specified herein.

5. PREPARATION FOR DELIVERY

5.1 PRESERVATION AND PACKAGING

5.1.1 LEVEL A. - Each vacuum pump shall be preserved in accordance with MIL-P-116, method II.

5.1.1.1 Prior to preservation, all fuel and oil shall be drained from the vacuum pump. All internal parts of the engine and vacuum pump which may be subject to corrosion shall be thoroughly slushed with a compound conforming to MIL-C-6529.

5.1.2 LEVEL C. - Vacuum pumps shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 PACKING

5.2.1 LEVEL A. - Vacuum pumps preserved and packaged as specified in 5.1.1 and 5.1.1.1 shall be packed in exterior shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or MIL-B-10377. Insofar as practicable, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 LEVEL B. - Vacuum pumps preserved and packaged as specified in 5.1.1 shall be packed in domestic exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior

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containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 LEVEL C. - Packages which require overpacking for acceptance by the carrier shall be packed in exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers, as applicable to the mode of transportation.

5.3 PHYSICAL PROTECTION. - Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100 except that waterproofing requirements for cushioning materials and containers shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 MARKING. - Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

VACUUM PUMP, ROTARY, POWER DRIVEN, 12 CFM WITH GASOLINE ENGINE, SKID MOUNTED, TYPE MA-1

6. NOTES

6.1 USE. - The Type MA-1 vacuum pump covered by this specification is intended primarily for use in evacuating the insulating annular space of liquid oxygen and nitrogen storage tanks.

6.2 ORDERING DATA. - Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Special tools to be furnished (See 3.12.2)

c. Whether tool compartment should be provided (See 3.12.3)

d. Applicable level of preparation for delivery (See section 5)

6.2.1 PREPRODUCTION TESTS. - One vacuum pump will be required as a preproduction sample to be subjected to the preproduction tests to determine compliance with the requirements of this specification. The point of inspection for these tests will be specified.

6.3 DEFINITION

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6.3.1 AMBIENT CONDITIONS. - Ambient conditions shall be understood to mean the dry-bulb temperature and the barometric pressure of the air approaching the pump inlet connection.

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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